

ANALYSTLAB AFRICA EXPERIENCE LAB

Official Week 7 Project Summary — Machine Learning Engineering Track

Intern Name: Grai Rudolf

Track: Machine Learning Engineering

Project Title: HealthConnect Clinic Appointment Attendance Optimization

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1. What I Planned to Test

The integrated ML pipeline built in Week 6 — its data-processing workflow, preprocessing-model interaction, expected and invalid inputs, expected outputs, reproducibility, and artifact persistence — to determine whether it performs reliably and is ready for Week 8 final integration.

2. What I Actually Tested

- Data-processing workflow (raw schema, cleaning, target, leakage).
- Preprocessing components (shape consistency, feature extraction).
- Preprocessing-model interaction (integration tests).
- Invalid/unexpected inputs (missing columns, null values, non-numeric threshold).
- Artifact save/load round-trip.
- Reproducibility (two independent runs with fixed seeds).

Total: **31 tests** (12 unit + 5 integration + 14 reliability), all passing.

3. Most Important Testing Results

- Valid records predict correctly with structured risk-tiered output.
- Reproducibility confirmed: two runs select the same best model (Logistic Regression).
- Artifact round-trip yields identical single-record predictions.
- Model metrics stable: recall 0.647, ROC-AUC 0.664 on held-out test.

4. Issues or Weaknesses Identified

1. `predict_single` with missing column raised a cryptic `KeyError` (High).
2. Null values in prediction input not guarded (Medium).
3. Decision threshold outside $[0, 1]$ silently accepted (Medium).
4. No `load_artifacts()` — save had no inverse (Medium).
5. No reproducibility verification (Low).

5. Improvements / Refinements Made

1. R1: Added `validate_prediction_input()` for single/batch prediction inputs.
2. R2: Added `validate_threshold()` guarding threshold range and type.
3. R3: Added `HealthConnectPipeline.load_artifacts()`.
4. R4: Added error handling around split, training, and prediction stages.
5. R5: Added `verify_reproducibility()`.

6. Retesting Results

All 14 new reliability tests pass post-refinement; the full suite of 31 tests is green. Malformed inputs now raise clear, actionable `ValidationError` messages.

7. Track(s) Collaborated With

Data Science track (HC-POD testing activity).

8. What Was Tested Collaboratively

Whether the integrated pipeline correctly consumes the Data Science candidate model (feature schema, probability output, risk-tier thresholds) and produces the agreed prediction output schema.

9. What Changed as a Result

Prediction input and output validation contracts were formalised; malformed records are rejected early with clear errors; the pipeline interface is now reliable and auditable.

10. Key Findings / Validation Outcomes

The pipeline correctly processes the documented inputs and produces valid, reproducible, risk-stratified predictions. It is ready to be wrapped behind an inference service in Week 8.

11. Major Challenges Encountered

Balancing a comprehensive test suite against runtime (31 tests, including two full pipeline runs for reproducibility); ensuring error messages are informative without leaking data.

12. Important Decisions Made

- Refinements raise `ValidationError` (fail-fast) rather than warnings.
- Logistic Regression retained as the reported best model for interpretability.
- Reproducibility verified via two full runs rather than a lighter check.

13. Remaining Limitations

- GroupKFold by `patient_id` not yet implemented.
- FastAPI/Docker deployment and drift monitoring deferred to Week 8.
- LightGBM tuned recall still below Random Forest.

14. Remaining Issues or Dependencies

Cross-track dependency on Data Science for final model sign-off; dependency on Data Analytics for validated feature findings before final integration.

15. Contribution to the Overall HealthConnect Project

The ML Engineering track now provides a validated, reproducible, and reliable pipeline that converts raw appointment data into risk-stratified predictions — the core technical foundation for the HealthConnect no-show intervention system.

16. What Must Be Completed Before Week 8

1. FastAPI inference service wrapping the validated pipeline.
2. GroupKFold final leakage control.
3. MLflow tracking and Evidently AI drift monitoring.
4. Final end-to-end validation across all HealthConnect components.